

■ ■ Blue Ridge Stream Restoration & Mitigation LLC — Enterprise Repository

Welcome to the central B2B operations repository for **Blue Ridge Stream Restoration & Mitigation LLC**. This workspace is designed for Co-Founder & Managing Director **Hunter Morris** and Board Member **Hadi Irvani** to lead North Georgia's coldwater stream restoration and compensatory mitigation credit market. This repository serves as a highly structured strategic, technical, and regulatory resource center, paired with a modern, responsive two-page web application designed for B2B credit buyers (hyperscale data centers, civil engineers, DOT planners) and private mountain landowners.

I. Core Philosophy: The Triple-Bank Semantic Framework

Traditional environmental conglomerates treat stream restoration as a sterile transaction, alienating rural families with complex regulatory jargon. Blue Ridge Stream Restoration overcomes this friction by leveraging the triple-meaning of the word **Bank** to unify ecology, regulation, and family wealth into a single storytelling pillar:

[THE TRIPLE-BANK FRAMEWORK]		
1. THE GEOMORPHIC BANK	2. THE REGULATORY BANK	3. THE WEALTH & LAND BANK
(Fluvial & Spawning Ecology)	(Savannah District Ledger)	(Landowner Family Legacy)
Vertical bank regrading	Savannah District IRT	70/30 Joint Venture split
to a stable 3:1 slope	accreditation, credit	yielding \$696k payout
with Rhododendron cover grades	deposits on RIBITS database	zero-cost streamside upgrades

- The Geomorphic Bank (Physical Stream Bed):** We repair vertical, eroding pasture banks, regrading them to stable 3:1 slopes and transplanting mature *Rhododendron maximum* to create thermal canopy shade, keeping water temperatures below 65°F for native trout spawning.
- The Regulatory Bank (Savannah District Credit Ledger):** We register the geomorphic improvements under USACE Savannah District guidelines, converting ecological restoration into certified, high-yield stream mitigation credits deposited in the federal RIBITS database.
- The Wealth & Land Bank (Landowner Financial Legacy):** We sell these credits to exurban commercial developers or hyperscale data center campuses (QTS, Microsoft, Google) at premium pricing. Under our 70/30 Joint Venture model, landowners receive a **30% cash payout split** (e.g. \$696,600 on our 1,500-LF Anderson Creek reach) or premium civil streamside upgrades constructed by Hunter's active 12-to-20+ field crew at zero out-of-pocket cost.

II. Repository Directory Structure (PhD-Grade Reorganization)

The repository has been programmatically sorted into 10 highly organized, color-coded subdirectories, aligning all strategic, technical, and legal assets into structured tiers:

ACR Stream Restoration/	
■■■ 01_Strategic_&_Business_Plans/	# Vision documents, recruiting plans, Traction
EOS scorecards	
■■■ 02_Regulatory_&_USACE_Frameworks/	# USACE banking rules, legal guidelines, bioeng
ineering manuals	
■■■ 03_Project_Case_Studies/	# Flagship geomorphic case studies and NWP 27 P
CN templates	
■■■ 04_Market_&_Competitor_Intelligence/	# Savannah District registries and exurban data
center leads	
■■■ 05_Sales_&_Outreach_Playbooks/	# Warm B2B outreach scripts, email structures,
and playbooks	
■■■ 06_Legal_&_Cooperative_Agreements/	# eSignature playbooks and land mitigation deed
covenants	
■■■ 07_Marketing_&_Social_Content/	# LinkedIn strategies, brand style guides, and
cinematic videos	
■■■ 08_Recruiting_&_Internships/	# 12 specialized UGA/Clemson/GT intern training
workstreams	
■■■ 09_Operations_&_Technical_Playbooks/	# Drone photogrammetry, RTK GNSS, and hosting s
etups	
■■■ 10_Outreach_&_Relationship_Banking/	# Ag/timber banker databases and B2B referrals
directories	
■■■ index.html	# Public Landowner Siting Page & Yield Calculat
or	
■■■ portal.html	# Private Operations & Strategic Resource Porta
l	
■■■ styles.css	# Global Trout-HSL Design Tokens & Layout style
sheet	

III. Web Application Architecture

The public-facing portal is constructed using a modern, performant **Vanilla HTML5, CSS3, and ES6 JavaScript** architecture, bypassing heavy Javascript frameworks for lightning-fast page loading and 100% SEO optimization.

1. Public Landowner Sourcing Page ([index.html](#))

Designed to capture exurban B2B credit requests and source high-yield rural stream tracts in North Georgia:

- **Trout-Stewardship Contrast:** Highlights Hunter Morris's niche as North Georgia's leading outfitter (directing a crew of 12 to 20+ active river professionals), contrasting geomorphic stream craftsmanship against corporate conglomerates that clear-cut and use heavy concrete rip-rap.
- **Tabs Case Studies Console:** Interactive, responsive CSS-tabs showcasing commercial stream credits (Roya's Cabin/Anderson Creek) vs trout habitat step-pool restoration (Hunter's Cabin/Goldmine Hollow).
- **Free Stream Restoration Yield Estimator:** Real-time Javascript calculator allowing landowners to enter their stream length and bank erosion severity to project stream credit yields, covered CapEx, and landowner gross cash payouts under the 70/30 split.

2. Private Operations & Strategic Resource Portal ([portal.html](#))

The secure, B2B central database for partners, lawyers, and relationship lenders:

- **Glassmorphic Folder Selector:** Responsive dashboard equipped with HSL trout-coordinated folder cards, custom vector SVG operational icons, and active indicator animations.
- **Dynamic Real-Time JavaScript Search & Filters:** A comprehensive, programmatic search algorithm that filters and renders all **72 enterprise files** in real time by operation tiers, subdirectories, titles, and descriptions.
- **High-Definition Widescreen Video Players:** Widescreen video players (1920x1080, 16:9 ratio) featuring slow-motion leaps of wild trout and slow-motion casting zooms, narrated in a deep, warm nature documentary style (David Attenborough tone) to establish high-end brand authority.

3. Styling Core ([styles.css](#))

Establishes the design tokens and variables used across all web layouts:

- **HSL Palette Tokens:** River Navy [hsl\(209, 59%, 29%\)](#), Trout Teal [hsl\(168, 54%, 37%\)](#), Copper Rust/Accent [hsl\(18, 63%, 43%\)](#), and Warm Gold [hsl\(43, 80%, 54%\)](#).
- **Responsive Flexbox/Grid:** Clean container grids with smooth transitions and glassmorphic micro-animations.

IV. Dynamic Calculations & Engineering Rigor

All geomorphic and financial calculators running in the background utilize verified fluvial geomorphology and Savannah District SOP equations:

1. Geomorphic Soil-Loss Equation

Calculates the exact tonnage of sediment prevented from choking local trout spawning gravels annually:

$$\text{Sediment Prevented (Tons/Year)} = (L * H * Re * pb) / 2000$$

Where:

- **L** = Stream reach length (Linear Feet)
- **H** = Stream bank height (Feet)
- **Re** = Lateral erosion rate (Feet/Year), measured using bank pins.
- **pb** = Soil bulk density constant ([85.0 lb/ft³](#) for North Georgia sand-clay matrices).
- **2000** = Conversion constant from pounds to short tons.

2. Boundary Shear Stress Equation

Ensures instream logs, root-wads, and J-hook structures remain rock-solid during 100-year storm events:

$$\text{Boundary Shear Stress } (\tau) = \gamma * R * S$$

Where:

- **τ** = Fluvial Boundary Shear Stress (lb/ft² or N/m²)
- **γ** = Unit weight of water (62.4 lb/ft³ or 9,810 N/m³)
- **R** = Hydraulic Radius (ft or m), calculated as cross-sectional bankfull area (A) divided by wetted perimeter (P).
- **S** = Channel slope gradient (ft/ft or m/m).

V. Quick GitHub Pages Deployment Guide

Hunter Morris can easily deploy and share this active B2B portal in under 2 minutes utilizing **GitHub Pages** for \$0.00 in hosting costs:

Step 1: Create a GitHub Repository

1. Log into GitHub at github.com.
2. Click **New** to create a new repository.
3. Name the repository: **ACR-Stream-Restoration**.
4. Choose **Public** or **Private** (GitHub Pages supports private repositories on Pro plans; select Public for free hosting).
5. Leave "Add a README" unchecked (we already have this master README).

Step 2: Push Local Files to GitHub

Open your terminal inside the local directory

/Users/irvani/Library/CloudStorage/GoogleDrive-hadi.irvani@gmail.com/My Drive/ACR Stream Restoration and execute:

```
# Initialize git (if not already done)
git init

# Add the remote GitHub origin (replace <your-username> with your actual GitHub username)
git remote add origin https://github.com/<your-username>/ACR-Stream-Restoration.git

# Set main branch
git branch -M main

# Push the committed files to GitHub
git push -u origin main
```

Step 3: Enable GitHub Pages

1. On your GitHub repository page, click on **Settings** (gear icon on the top menu).
2. On the left sidebar, under the "Code and automation" section, click on **Pages**.
3. Under the **Build and deployment** section, select **Deploy from a branch** as the Source.
4. Under the **Branch** dropdown, select **main** and set the folder to **/ (root)**.
5. Click **Save**.

Step 4: Access and Share the Portal!

GitHub will generate a live, secure URL for your venture within 60 seconds. The URL will follow this structure:

■ **<https://<your-username>.github.io/ACR-Stream-Restoration/>**

You can immediately copy and send this link to:

- **Hunter Morris** to view the dynamic dashboard.
- **Private Landowners** to estimate their geomorphic credit split payouts.
- **B2B Buyers & Developers** to check active credit inventories and download CWA Section 404 compliance guidelines.

VI. Team & Academic Leveraged Staffing model

Blue Ridge Stream Restoration operates at hyper-scale speeds using an **Academic-Leveraged model**: achieving a 10-person capacity firm with only 3 Full-Time Equivalents (FTEs). We rotate specialized graduate interns to run our core technical pipelines:

- **University of Georgia (UGA) Warnell School of Forestry**: Runs field biology, EPA rapid bioassessments, HOBO water probes, and vegetative buffer surveying.
- **Georgia Institute of Technology**: Formulates 1D/2D HEC-RAS hydraulic models, processes RTK survey thalwegs, and drafts AutoCAD Civil 3D bioengineering construction plans.
- **Clemson University MRED Program**: Handles real estate NPV underwriting, records conservation easements, and audits escrow balances.

Developed by Hunter Morris & Hadi Irvani. Built with geomorphic and operational integrity.